

Sure! Here's a detailed answer to each of the questions you listed. These are structured clearly and concisely to match exam or assignment expectations for topics from **Software Project Management** or **Software Engineering** subjects.

Q1) a) What is a project? Why is software project management important? [5 Marks]

Project:

A **project** is a temporary endeavor undertaken to create a unique product, service, or result. It has a **specific goal, defined timeline, allocated resources**, and **constraints** such as scope, cost, and quality.

Software Project Management Importance:

Software project management (SPM) is the discipline of planning, organizing, leading, and controlling software projects. It is important because:

1. **Efficient Planning:** Helps in defining scope, timeline, and budget.
 2. **Resource Management:** Allocates the right people and tools effectively.
 3. **Risk Management:** Identifies and mitigates risks early.
 4. **Quality Assurance:** Ensures the final product meets requirements.
 5. **Meeting Deadlines and Budget:** Reduces cost and time overruns.
-

Q1) b) How do plans, methods, and methodologies differ from each other? [5 Marks]

Aspect	Plan	Method	Methodology
Definition	A roadmap or strategy for achieving objectives.	A procedure or technique used to perform tasks.	A collection of methods and best practices.
Scope	Project-specific.	Task-specific.	Project-wide, framework-level.
Example	Project timeline or Gantt chart.	Code review, testing.	Agile, Waterfall, Scrum.
Purpose	To define what, when, and how things happen.	To perform specific tasks or solve problems.	To guide the entire software development lifecycle.

Q1) c) Describe contract management in detail. [5 Marks]

Contract Management is the process of managing contracts made with customers, vendors, partners, or employees. It includes:

1. **Contract Creation:** Drafting and agreeing on the contract terms (cost, deliverables, time).
2. **Contract Execution:** Monitoring deliverables, ensuring compliance.

3. **Monitoring Performance:** Tracking milestones, quality, and payments.
4. **Change Management:** Handling scope changes through amendments.
5. **Closure:** Formal acceptance and evaluation post-delivery.

Effective contract management ensures that both parties fulfill their obligations and minimizes risks of disputes or legal issues.

OR (Alternative to Q1 c)

Q2) a) Identify the management responsibilities of the manager in view of software project management. [5 Marks]

A **Software Project Manager** has multiple responsibilities:

1. **Project Planning:** Define scope, schedule, and resources.
 2. **Team Management:** Assign tasks, resolve conflicts, motivate team.
 3. **Risk Management:** Identify, assess, and mitigate risks.
 4. **Quality Assurance:** Ensure deliverables meet quality standards.
 5. **Monitoring & Reporting:** Track progress, prepare status reports.
 6. **Stakeholder Communication:** Liaise with clients and stakeholders.
-

Q2) b) Explain traditional project management and modern project management. [5 Marks]

Aspect	Traditional Project Management	Modern Project Management
Approach	Linear (Waterfall model)	Iterative (Agile, Scrum, Kanban)
Flexibility	Less flexible to changes	Highly adaptable to changes
Planning	Heavy upfront planning	Continuous planning throughout
Client Involvement	Minimal, usually at start and end	High involvement throughout development
Delivery	Delivered at the end	Delivered in iterations
Tools Used	Gantt charts, PERT, MS Project	Jira, Trello, Agile boards

Q2) c) Define business case and explain the concept of business case. [5 Marks]

Definition:

A **business case** is a document that justifies the initiation of a project. It outlines the **benefits**, **costs**, **risks**, and **alternatives** involved.

Concept:

- Helps decision-makers determine whether the project is worth pursuing.
- Includes:
 - Problem Statement
 - Solution Overview
 - Cost-Benefit Analysis
 - Risks and Mitigations
 - Return on Investment (ROI)

A good business case ensures alignment with business goals and secures executive approval.

Q3) a) Draw the activity diagram in reference to online shopping system. [5 Marks]

Here is a simple **activity diagram** (text version) for an online shopping system:

```
Start
|
Browse Products
|
[Decision] -> Add to Cart?
|Yes          |No
Add to Cart    Browse More
|
Proceed to Checkout
|
Enter Shipping Details
|
Make Payment
|
Receive Order Confirmation
|
End
```

You can draw this in UML with appropriate symbols for decision nodes, activity states, start/end nodes, and transitions.

Q3) b) Explain GQM Paradigm. [5 Marks]**GQM: Goal – Question – Metric**

A structured approach for defining and interpreting software measurements.

1. **Goal:** Define what you want to achieve (e.g., improve code quality).
2. **Question:** Develop questions to understand if goals are met (e.g., How many bugs are found in testing?).
3. **Metric:** Identify metrics to answer the questions (e.g., number of defects per 1,000 LOC).

GQM helps organizations align software metrics with business objectives.

Q3) c) Enlist the techniques of process analysis and explain in brief. [5 Marks]

Techniques:

1. **Root Cause Analysis (RCA):** Identifies underlying causes of problems.
2. **Process Mapping:** Visual diagrams of workflows (e.g., flowcharts).
3. **Gap Analysis:** Compares current vs. desired performance.
4. **SWOT Analysis:** Assesses strengths, weaknesses, opportunities, threats.
5. **Value Stream Mapping:** Focuses on value-added vs. non-value-added steps.

These techniques help in identifying inefficiencies and improving processes.

OR (Alternative to Q3)

Q4) a) Draw the use case diagram in reference to online shopping system. [5 Marks]

Actors:

- Customer
- Admin

Use Cases:

- Register/Login
- Browse Products
- Add to Cart
- Checkout
- Make Payment
- Track Order
- Manage Products (Admin)
- View Orders (Admin)

You can draw this in UML with:

- Ovals for use cases
 - Stick figures for actors
 - Lines showing associations.
-

Q4) b) What is project evaluation? Explain its importance. [5 Marks]

Project Evaluation is the process of assessing a project's performance against objectives, in terms of cost, time, quality, and impact.

Importance:

1. Measures success and ROI.
 2. Identifies lessons learned.
 3. Helps improve future projects.
 4. Ensures accountability and transparency.
 5. Facilitates decision-making for stakeholders.
-

Q4) c) Describe "Return on Investment" cost-benefits evaluation technique with example. [5 Marks]

ROI (Return on Investment):

Measures the profitability of a project.

Formula:

$$ROI = \frac{\text{Net Profit}}{\text{Total Investment}} \times 100$$

Example:

- Total cost of project = \$50,000
- Benefit gained = \$70,000
- Net Profit = \$70,000 - \$50,000 = \$20,000

$$ROI = \frac{20,000}{50,000} \times 100 = 40\%$$

Conclusion: The project has a 40% ROI, meaning it's profitable and worth the investment.

Let me know if you want **diagrams in image format** or a **PDF document** with all answers neatly formatted.